Referee report: evolving reputation for commitment

Summary:

This paper makes an important contribution to the literature on monetary policy games. This is an old literature that goes back to Barro-Gordon (1983), Rogoff (1985) etc. This paper makes two key contributions to previous work: first, it introduces reputation as a state variable in a dynamic stochastic environment with heterogeneous types and asymmetric information; second, it develops a framework based on reputation that is ready for serious quantification, which is unlike much other work in the literature. To that end, the authors develop a new recursive solution method.

In the model the authors develop, a policymaker (CB, central bank) plays against agents in the economy (firms, say). The CB can have two types, committed (~ Ramsey) and opportunistic (~ myopic Markov). The CB is replaced at a Poisson rate (common knowledge) and the type is redrawn. While the CB stays, agents learn about the type of the CB based on its actions. The posterior over the type is what we will call "reputation". As in much of the literature on monetary policy games, high reputation means that agents think the CB is committed. From the CB's perspective, high reputation is desirable because agents give it a better inflation-output tradeoff (via endogenous inflation expectations). As a result, an opportunistic type will have an incentive to pool with the committed type to some extent. What's critical is the authors' assumption that agents observe neither the CB's type nor her action; that is, agents only observe the CB's actions with some noise (implementation error). This gives the opportunistic type some wiggle room.

The paper characterizes the PBE of this game and develops a recursive representation. This recursive representation spans Propositions 1 and 2 and is the main analytical contribution of the paper. After its characterization, the paper embarks on a quantitative and empirical analysis of the calibrated model. There are three parts to this second half of the paper: (i) taking the model to the data via a …, (ii) rationalizing the Volcker disinflation within the model, and (iii) validating the model using regression analysis of longer-term inflation expectations.

Comments:

1. From my own experience playing around with models of monetary policy games with reputation, I've learned that there is a large menu of possible modeling choices, including commitment technology, preferences, etc. And these choices have really significant implications for the qualitative features of the model and for what exactly reputation does. Here, the authors opt for a two-type representation where type 1 has perfect commitment and type 2 is not only Markov but also myopic. The authors justify this

choice by appealing to the model's "success in matching US experience of inflation and inflation expectations". I am very sympathetic to this approach. But I believe the authors could increase their impact by making their exposition align more closely with this approach. At the moment, the paper reads (to me) as follows: *We introduce a model, we study this model, and then somewhat as an afterthought we show that the model is consistent with empirical regularities about US historical inflation episodes.* (This is admittedly a little exaggerated in the interest of clarifying my comment.) I think a structure and exposition along the following lines would be more successful: *Here is a set of empirical regularities that we believe are critical to understand (historical) episodes of inflation in the US. Some of these are known from the prior literature, some of them are new. (Hopefully, there would be strictly more than 2 such empirical regularities.) Previous models used in the literature to study these episodes cannot jointly explain these empirical regularities. We develop and quantify a new model based on a monetary policy game with reputation that allows us to jointly match and rationalize the empirical regularities.* While this suggestion represents largely a change in expositional structure and emphasis, I do believe the paper would benefit substantively from a more in-depth (and perhaps front-loaded) discussion and study of the relevant empirical regularities (and of why it takes a new model to make sense of them). My next comments discuss in more detail the empirical regularities that the paper currently addresses.

2. The most compelling result of the paper in my view is the following: The authors show that a "naïve" policymaker who ignores the impact of policy on reputation "takes significantly longer to disinflate the economy than what was observed during the Volcker disinflation following 1981". My optimistic interpretation of this result would be: The standard NK model cannot rationalize the Volcker disinflation. That's because optimal monetary policy in the standard NK model is concerned with the inflation-output tradeoff; but policy affects neither reputation nor, at least in the simple model, longer-run inflation expectations. My question is the following: Is it true that the standard NK framework struggles in a deep way with rationalizing the Volcker period? If this were true, it would be a very compelling motivation for the paper. And I would start with that in paragraph 1. But my sense is that the answer to this question is a little more nuanced, and there are extensions of the simple NK model that could have more success at rationalizing the Volcker episode (including extensions that endogenize longer-run expectations and solve the model non-linearly). But in my reading, this is important for the framing of this paper. At the moment, the motivational paragraph 1 simply asserts that expectation management is an important dimension of monetary policy. But I would find this motivation more compelling if the paper pointed to a set of empirical regularities that the current class of models used for policy analysis is unable to match and rationalize.

3. I found the discussion in Section 6.3 a little less clear and accessible than the rest of the paper. But I am also not particularly familiar with that literature. In particular, I did not fully appreciate why regression Model 3 was uniquely associated with the model presented in this paper. Is this a formal statement? In any case, if the authors believe that the empirical regularity they identify in Section 6.3 is novel and important, I might consider emphasizing it more and earlier (in the spirit of my first comment). Is this an empirical regularity that other models (including models that use alternative approaches to endogenous inflation expectations) cannot explain?

4. The paper proposes to explain the history of US inflation since 1970 with only cost-push shocks. What are the model's quantitative implications for economic activity (output gap, unemployment, or equivalent)? Shouldn't the absence of demand shocks complicate identification especially after Volcker, where demand shocks seem to have predominant?

5. The paper makes several stark assumptions. These include the assumption of a myopic Markov type, as well as the absence of any public signals that would help discipline the opportunistic type. As a matter of applied theory, it seems important to explore robustness around these. The authors frame their contribution as a quantitative one, which I agree with. But then it becomes even more critical to emphasize the empirical regularities that the model manages to rationalize despite these stark assumptions. From a quantitative perspective, the assumption that agents observe neither the CB's type nor her actions seems like an appropriate description of central banking in the 1970s, but less so today. The authors' assumption that agents only observe the CB's action with noise might be interpreted as a reduced-form expression of the fact that the CB has private information, for which there is a lot of empirical evidence. So even though the interest rate as well as many macro indicators are public knowledge, the CB sets the interest rate as a function of the "state of the economy", which is arguably at least in part private information. After Greenspan, however, the Fed made a big effort to increase transparency. My understanding is that empirical evidence still points to private information today, but it would be difficult to argue in this context that the same model can be used to understand the inflation episodes of the 1970s and 2021.

A good summary of my suggestions would be this: I am very enthusiastic about the broader direction to rationalize US monetary time series in the context of endogenous reputation models. And I think the present paper makes an important step in that direction, by presenting a benchmark model that is ready for quantitative analysis. At the same time, I am hopeful to gain further conviction that the model the authors present is the *right* model to explain monetary policy. And this comes down to identifying and documenting the key empirical regularities, and then showing that the proposed model can rationalize them in a way that other models cannot.